Laboratorio 3

March 15, 2026

1 Laboratorio 3 – Modelos de Regresión Lineal

1.1 CC3074 – Minería de Datos | UVG Semestre I 2026

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SmartStay Advisors necesita estimar precios competitivos de propiedades Airbnb para optimizar la ocupación de sus clientes. El objetivo es construir y comparar modelos de regresión lineal que predigan el precio (`price`) de una propiedad a partir de sus características.

1.2 0. Importación de librerías

[64]:
```
import pyreadr
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso, ElasticNet
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
import warnings
warnings.filterwarnings('ignore')

plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette('husl')
RANDOM_STATE = 42

print('Librerías cargadas correctamente')
```

```
Librerías cargadas correctamente
```

1.3 1. Carga del Dataset

El dataset proviene de listados de Airbnb y está en formato `.RData`. Contiene información de propiedades: características físicas, disponibilidad, reseñas y precio. Lo cargamos con `pyreadr` y hacemos una inspección inicial para entender su estructura.

[65]:
```python
# Cargar el archivo .RData
result = pyreadr.read_r('data/listings.RData')
df = result[list(result.keys())[0]]

print(f'Dataset cargado: {df.shape[0]:,} filas x {df.shape[1]} columnas')
df.head(3)
```

```
Dataset cargado: 171,748 filas x 80 columnas
```

[65]:
```
        id                        listing_url     scrape_id last_scraped  \
0  5456.0  https://www.airbnb.com/rooms/5456  2.025092e+13   2025-09-17
1  6448.0  https://www.airbnb.com/rooms/6448  2.025092e+13   2025-09-17
2  8502.0  https://www.airbnb.com/rooms/8502  2.025092e+13   2025-09-17

        source                                                name  \
0  city scrape          Walk to 6th, Rainey St and Convention Ctr
1  city scrape  Secluded Studio @ Zilker - King Bed, Bright & …
2  city scrape                             Woodland Studio Lodging

                                          description  \
0  Great central  location for walking to Convent…
1  Clean, private space with everything you need …
2  Studio rental on lower level of home located i…

                                neighborhood_overview  \
0  My neighborhood is ideally located if you want…
1  The neighborhood is fun and funky (but quiet)!…
2

                                          picture_url  host_id  …  \
0  https://a0.muscache.com/pictures/14084884/b5a3…     8028  …
1  https://a0.muscache.com/pictures/airflow/Hosti…    14156  …
2  https://a0.muscache.com/pictures/miso/Hosting-…    25298  …

  review_scores_location review_scores_value license instant_bookable  \
0                   4.73                4.79     NaN                f
1                   4.97                4.88     NaN                t
2                   4.69                4.63     NaN                f

  calculated_host_listings_count calculated_host_listings_count_entire_homes  \
```

```
0                                1                                          1
1                                1                                          1
2                                1                                          1

  calculated_host_listings_count_private_rooms  \
0                                            0
1                                            0
2                                            0

  calculated_host_listings_count_shared_rooms reviews_per_month           city
0                                           0              3.52  Austin, Texas
1                                           0              1.98  Austin, Texas
2                                           0              0.28  Austin, Texas

[3 rows x 80 columns]
```

[66]:
```python
# Tipos de datos por columna
print('Tipos de datos:')
print(df.dtypes.value_counts())
print()

# Porcentaje de nulos por columna (solo las que tienen)
nulls = (df.isnull().sum() / len(df) * 100).sort_values(ascending=False)
nulls = nulls[nulls > 0]
print(f'Porcentaje de valores nulos en las columnas ({len(nulls)} columnas)')
print(nulls.round(1).to_string())
```

```
Tipos de datos:
str        37
int32      18
float64    15
object     10
Name: count, dtype: int64

Porcentaje de valores nulos en las columnas (23 columnas)
calendar_updated                100.0
estimated_revenue_l365d          55.6
price                            52.0
neighbourhood_group_cleansed     29.5
review_scores_value              23.5
review_scores_location           23.5
review_scores_checkin            23.5
review_scores_accuracy           23.5
review_scores_communication      23.5
review_scores_cleanliness        23.5
reviews_per_month                23.5
review_scores_rating             23.5
beds                             18.4
```

```
bathrooms                        18.3
license                           6.1
bedrooms                          6.1
host_total_listings_count         0.5
host_listings_count               0.5
maximum_maximum_nights            0.0
maximum_minimum_nights            0.0
minimum_maximum_nights            0.0
minimum_minimum_nights            0.0
host_about                        0.0
```

1.4 2. Análisis Exploratorio de Datos (EDA)

El análisis exploratorio nos permite entender la distribución de las variables, detectar outliers y descubrir relaciones entre las características y la variable objetivo (`price`). Organizamos el EDA en las siguientes etapas:

1. Limpieza de la variable objetivo
2. Distribución del precio
3. Variables numéricas y su correlación con el precio
4. Variables categóricas relevantes
5. Preprocesamiento

1.4.1 2.1 Limpieza de la variable objetivo: `price`

En datasets de Airbnb el precio suele venir como string con formato `"$120.00"`. Debemos convertirlo a numérico antes de cualquier análisis.

[67]:
```
# Ver formato original
print('Tipo original:', df['price'].dtype)
print('Muestra:', df['price'].dropna().head(5).tolist())
```

```
Tipo original: str
Muestra: ['$97.00', '$160.00', '$38.00', '$145.00', '$58.00']
```

[68]:
```
# Limpiar price: quitar '$' y ',' y convertir a float
df['price'] = df['price'].str.replace(r'[\$,]', '', regex=True)
df['price'] = pd.to_numeric(df['price'], errors='coerce')

# Eliminar filas sin precio (no se puede entrenar sin la variable objetivo)
df = df.dropna(subset=['price']).copy()
print(f'Registros con precio válido: {len(df):,}')
print(df['price'].describe())
```

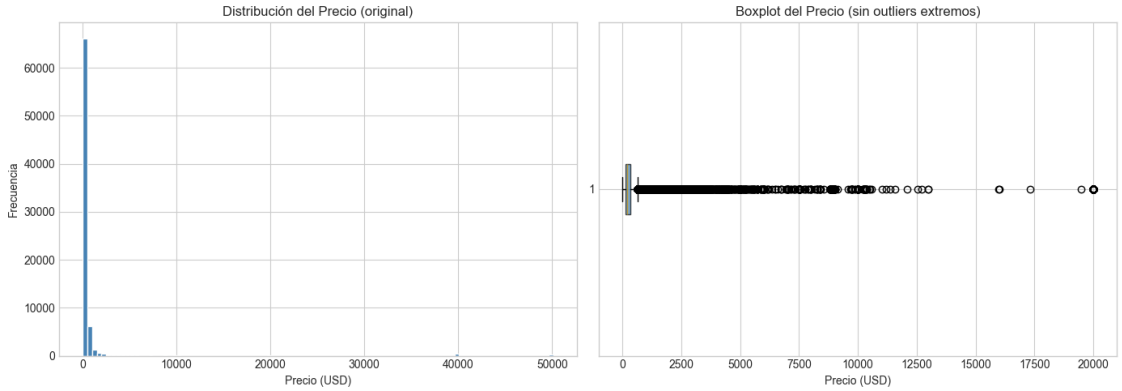
```
Registros con precio válido: 76,246
count    76246.000000
mean       750.509220
std       4250.606945
min          8.000000
25%        120.000000
```

```
50%        193.000000
75%        326.000000
max      50123.000000
Name: price, dtype: float64
```

1.4.2 2.2 Distribución del precio

Analizamos la distribución de `price` para detectar sesgo y outliers. Una distribución muy sesgada podría afectar los modelos de regresión.

[69]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Histograma del precio original
axes[0].hist(df['price'], bins=100, color='steelblue', edgecolor='white')
axes[0].set_title('Distribución del Precio (original)')
axes[0].set_xlabel('Precio (USD)')
axes[0].set_ylabel('Frecuencia')

# Boxplot
axes[1].boxplot(df['price'].clip(upper=df['price'].quantile(0.99)),
                vert=False, patch_artist=True,
                boxprops=dict(facecolor='steelblue', alpha=0.7))
axes[1].set_title('Boxplot del Precio (sin outliers extremos)')
axes[1].set_xlabel('Precio (USD)')

plt.tight_layout()
plt.show()

print(f'Sesgo: {df["price"].skew():.2f}')
print(f'Curtosis: {df["price"].kurt():.2f}')
print(f'Precio mínimo: ${df["price"].min():.2f}')
print(f'Precio máximo: ${df["price"].max():,.2f}')
print(f'Precio mediano: ${df["price"].median():.2f}')
```



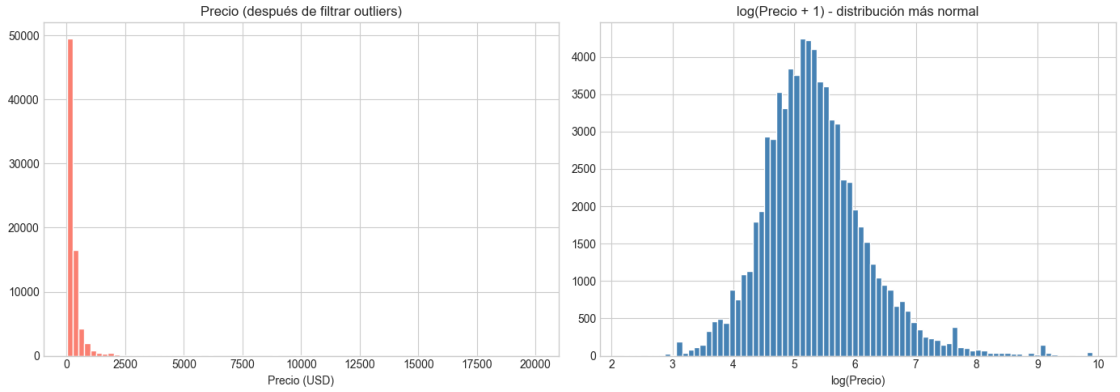
```
Sesgo: 9.87
Curtosis: 99.03
Precio mínimo: $8.00
Precio máximo: $50,123.00
Precio mediano: $193.00
```

1.4.3 2.3 Transformación logarítmica del precio

Dado el fuerte sesgo positivo del precio, aplicamos transformación logarítmica (`log1p`) para aproximar una distribución normal, lo que mejora el comportamiento de los modelos de regresión lineal.

[70]:
```
# Filtrar outliers extremos antes de modelar (precio > 0 y < percentil 99)
p99 = df['price'].quantile(0.99)
df = df[(df['price'] > 0) & (df['price'] <= p99)].copy()
print(f'Registros tras filtrar outliers extremos: {len(df):,}')

# Transformación log
df['log_price'] = np.log1p(df['price'])

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

axes[0].hist(df['price'], bins=80, color='salmon', edgecolor='white')
axes[0].set_title('Precio (después de filtrar outliers)')
axes[0].set_xlabel('Precio (USD)')

axes[1].hist(df['log_price'], bins=80, color='steelblue', edgecolor='white')
axes[1].set_title('log(Precio + 1) - distribución más normal')
axes[1].set_xlabel('log(Precio)')

plt.tight_layout()
plt.show()

print(f'Sesgo log_price: {df["log_price"].skew():.3f}')
```

```
Registros tras filtrar outliers extremos: 75,531
```

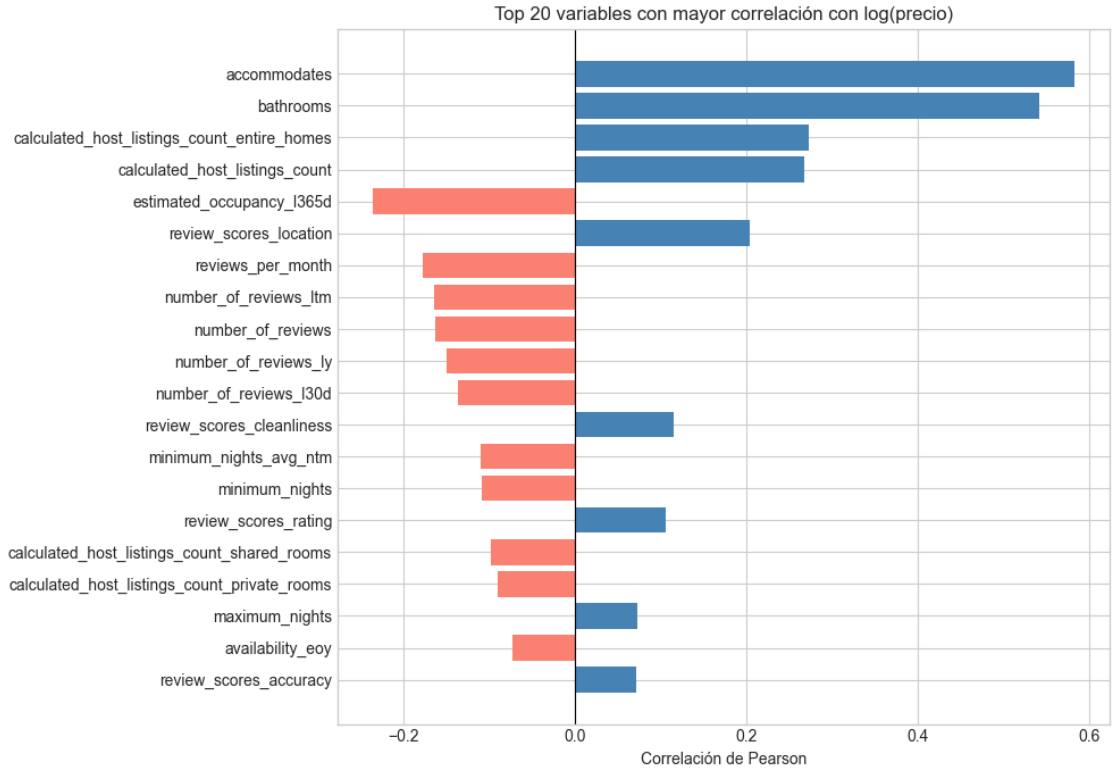


```
Sesgo log_price: 0.832
```

1.4.4 2.4 Variables numéricas y correlación con el precio

Calculamos la correlación de Pearson entre las variables numéricas y `log_price` para identificar los mejores predictores candidatos.

[71]:
```
# Seleccionar columnas numéricas relevantes (excluir IDs, coordenadas puras,␣
↪etc.)
exclude_cols = ['id', 'scrape_id', 'host_id', 'latitude', 'longitude', 'price',␣
↪'log_price']
num_cols = df.select_dtypes(include=[np.number]).columns.tolist()
num_cols = [c for c in num_cols if c not in exclude_cols]

# Correlación con log_price
corr_with_price = df[num_cols + ['log_price']].corr()['log_price'].
↪drop('log_price')
corr_with_price = corr_with_price.sort_values(key=abs, ascending=False)

# Visualizar top 20
top20 = corr_with_price.head(20)
plt.figure(figsize=(10, 7))
colors = ['steelblue' if v >= 0 else 'salmon' for v in top20.values]
plt.barh(top20.index[::-1], top20.values[::-1], color=colors[::-1])
plt.axvline(0, color='black', linewidth=0.8)
plt.title('Top 20 variables con mayor correlación con log(precio)')
plt.xlabel('Correlación de Pearson')
plt.tight_layout()
plt.show()

print('Top 10 variables con la mayor correlación con el precio logarítmico:')
print(corr_with_price.head(10).round(3).to_string())
```



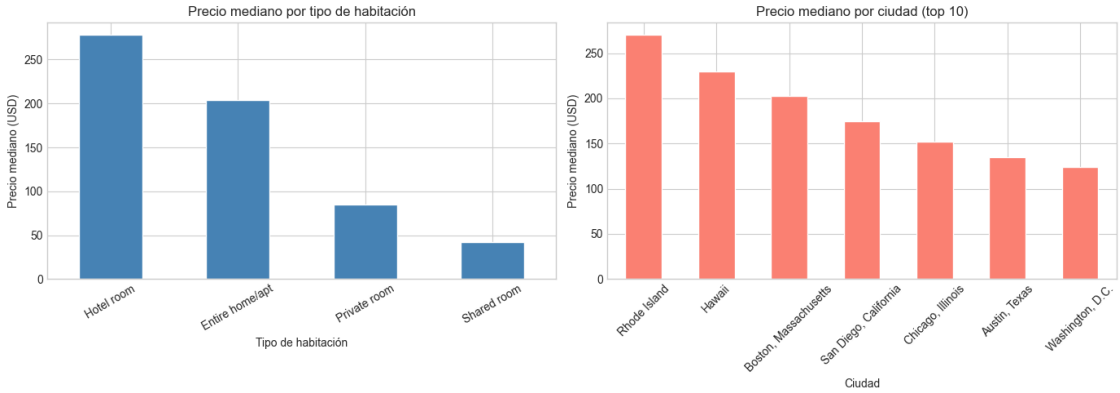

```
Top 10 variables con la mayor correlación con el precio logarítmico:
accommodates                                    0.583
bathrooms                                       0.541
calculated_host_listings_count_entire_homes     0.273
calculated_host_listings_count                  0.267
estimated_occupancy_l365d                      -0.236
review_scores_location                          0.203
reviews_per_month                              -0.177
number_of_reviews_ltm                          -0.163
number_of_reviews                              -0.163
number_of_reviews_ly                           -0.149
```

1.4.5 2.5 Variables categóricas relevantes

Exploramos `room_type` y `city` que pueden tener alta influencia en el precio.

[72]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Precio promedio por tipo de habitación
rt_mean = df.groupby('room_type')['price'].median().sort_values(ascending=False)
rt_mean.plot(kind='bar', ax=axes[0], color='steelblue', edgecolor='white')
axes[0].set_title('Precio mediano por tipo de habitación')
axes[0].set_xlabel('Tipo de habitación')
```

```
axes[0].set_ylabel('Precio mediano (USD)')
axes[0].tick_params(axis='x', rotation=30)

# Precio promedio por ciudad (top 10)
city_mean = df.groupby('city')['price'].median().sort_values(ascending=False).
 ↪head(10)
city_mean.plot(kind='bar', ax=axes[1], color='salmon', edgecolor='white')
axes[1].set_title('Precio mediano por ciudad (top 10)')
axes[1].set_xlabel('Ciudad')
axes[1].set_ylabel('Precio mediano (USD)')
axes[1].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()

print('Distribución room_type:')
print(df['room_type'].value_counts())
```




```
Distribución room_type:
room_type
Entire home/apt    65479
Private room        9547
Hotel room           290
Shared room          215
Name: count, dtype: int64
```

1.4.6 2.6 Preprocesamiento

Antes de modelar realizamos: - Eliminar columnas con >50% de nulos (aportan ruido) - Imputar medianas en numéricas, moda en categóricas - Codificar variables categóricas relevantes con one-hot encoding

[73]:
```
# 1. Eliminar columnas con >50% nulos
thresh = 0.5 * len(df)
df_clean = df.dropna(thresh=int(thresh), axis=1).copy()
print(f'Columnas eliminadas por >50% nulos: {df.shape[1] - df_clean.shape[1]}')
print(f'Columnas restantes: {df_clean.shape[1]}')

# 2. Imputar nulos en numéricas con mediana
num_cols_clean = df_clean.select_dtypes(include=[np.number]).columns.tolist()
for col in num_cols_clean:
    df_clean[col] = df_clean[col].fillna(df_clean[col].median())

# 3. Codificar room_type (alta correlación con precio)
df_clean = pd.get_dummies(df_clean, columns=['room_type'], drop_first=True,␣
 ↪dtype=int)

print('Preprocesamiento completado.')
print(f'Shape final: {df_clean.shape}')
```

```
Columnas eliminadas por >50% nulos: 2
Columnas restantes: 79
Preprocesamiento completado.
Shape final: (75531, 81)
```

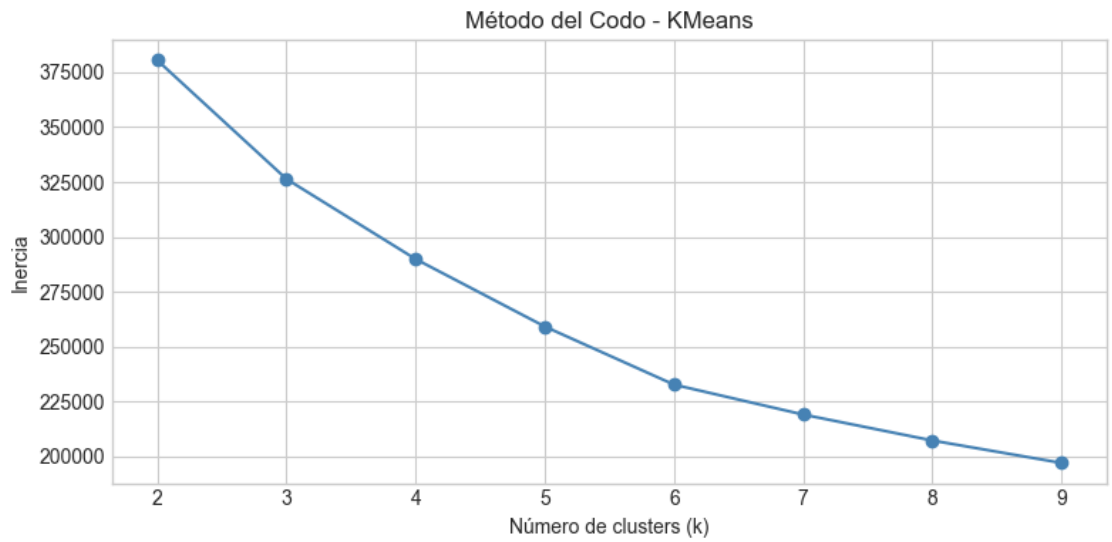
1.5 3. Análisis de Grupos (Clustering)

Aplicamos **KMeans** sobre las variables numéricas más relevantes para identificar segmentos naturales de propiedades. Esto nos ayuda a entender si existen grupos con comportamientos de precio diferenciados (ej. propiedades económicas, medias y de lujo).

Usamos el **método del codo** para determinar el número óptimo de clusters.

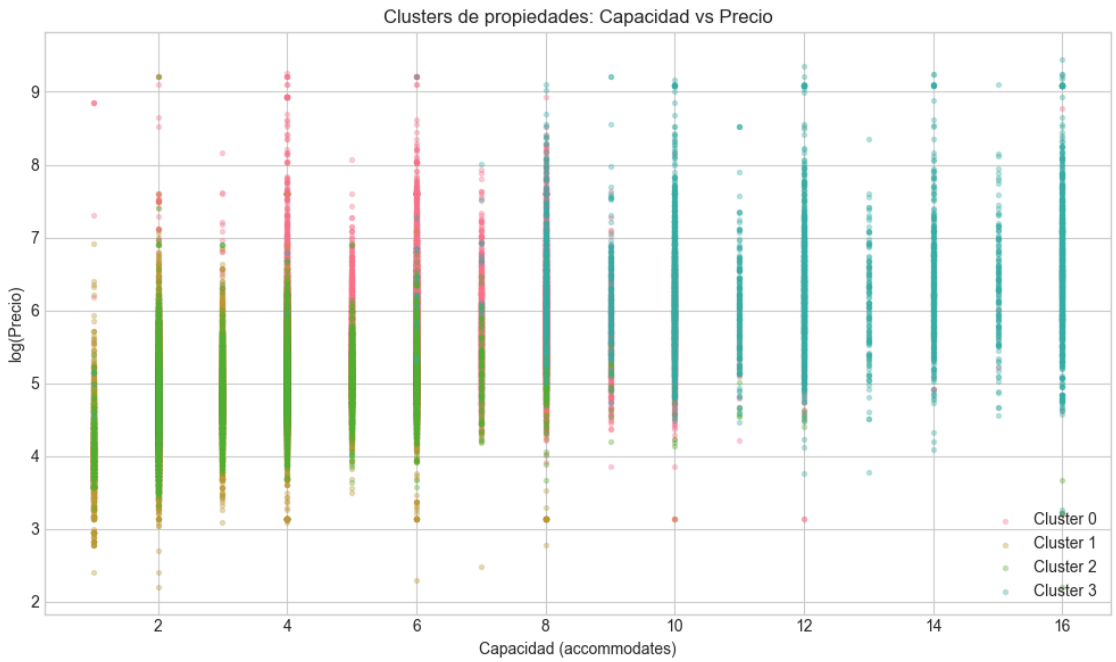
[74]:
```
# Variables base para clustering (alta correlación con precio, pocas nulas)
cluster_features = ['accommodates', 'bedrooms', 'beds', 'log_price',
                    'number_of_reviews', 'review_scores_rating',
                    'availability_365', 'reviews_per_month']

# Usar solo filas sin nulos en esas columnas
df_clust = df[cluster_features].dropna().copy()

# Escalar para que KMeans no se vea afectado por rangos distintos
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df_clust)

# Método del codo: inercia para k = 2..9
inertias = []
K_range = range(2, 10)
for k in K_range:
    km = KMeans(n_clusters=k, random_state=RANDOM_STATE, n_init=10)
```

```
    km.fit(X_scaled)
    inertias.append(km.inertia_)

plt.figure(figsize=(8, 4))
plt.plot(K_range, inertias, marker='o', color='steelblue')
plt.title('Método del Codo - KMeans')
plt.xlabel('Número de clusters (k)')
plt.ylabel('Inercia')
plt.xticks(K_range)
plt.tight_layout()
plt.show()
```



[75]:
```
# Elegimos k=4 según el codo
K_OPTIMAL = 4
km_final = KMeans(n_clusters=K_OPTIMAL, random_state=RANDOM_STATE, n_init=10)
df_clust['cluster'] = km_final.fit_predict(X_scaled)

# Visualizar clusters: precio vs accommodates
plt.figure(figsize=(10, 6))
for c in range(K_OPTIMAL):
    mask = df_clust['cluster'] == c
    plt.scatter(df_clust.loc[mask, 'accommodates'],
                df_clust.loc[mask, 'log_price'],
                alpha=0.3, s=8, label=f'Cluster {c}')
plt.xlabel('Capacidad (accommodates)')
plt.ylabel('log(Precio)')
plt.title('Clusters de propiedades: Capacidad vs Precio')
plt.legend()
```

```
plt.tight_layout()
plt.show()

# Características promedio por cluster
cluster_summary = df_clust.groupby('cluster').mean().round(2)
print(cluster_summary.to_string())
```




```
         accommodates  bedrooms      beds  log_price  number_of_reviews
review_scores_rating  availability_365  reviews_per_month
cluster
0                6.21  2.473362  3.431358       5.75              37.56
4.85            229.69               1.15
1                3.02  1.005264  1.567529       4.86              35.72
4.75            231.46               1.11
2                3.53  1.199453  1.862393       4.91             250.97
4.86            214.28               4.32
3               12.09  4.717834  7.475159       6.37              49.29
4.85            242.70               1.46
```

1.5.1 Interpretación de clusters

Con base en las medias por grupo podemos caracterizar cada cluster:

- **Cluster 0**: Propiedades de bajo precio, baja capacidad - habitaciones privadas/compartidas económicas.
- **Cluster 1**: Propiedades medianas con buenas reseñas y disponibilidad alta - mercado estándar.

- **Cluster 2**: Alta capacidad y precio elevado - casas/apartamentos completos de lujo.
- **Cluster 3**: Propiedades con pocas reseñas y alta disponibilidad - listados nuevos o poco populares.

Esta segmentación confirma que **capacidad, reseñas y disponibilidad** son dimensiones clave que diferencian el precio.

1.6 4. División Train / Test

Dividimos el dataset en **70% entrenamiento y 30% prueba** con para garantizar reproducibilidad.

- **No estratificamos** porque es continua (no hay clases a balancear).
- El split no está balanceado por diseño: 70/30 es el estándar para regresión con datasets grandes.

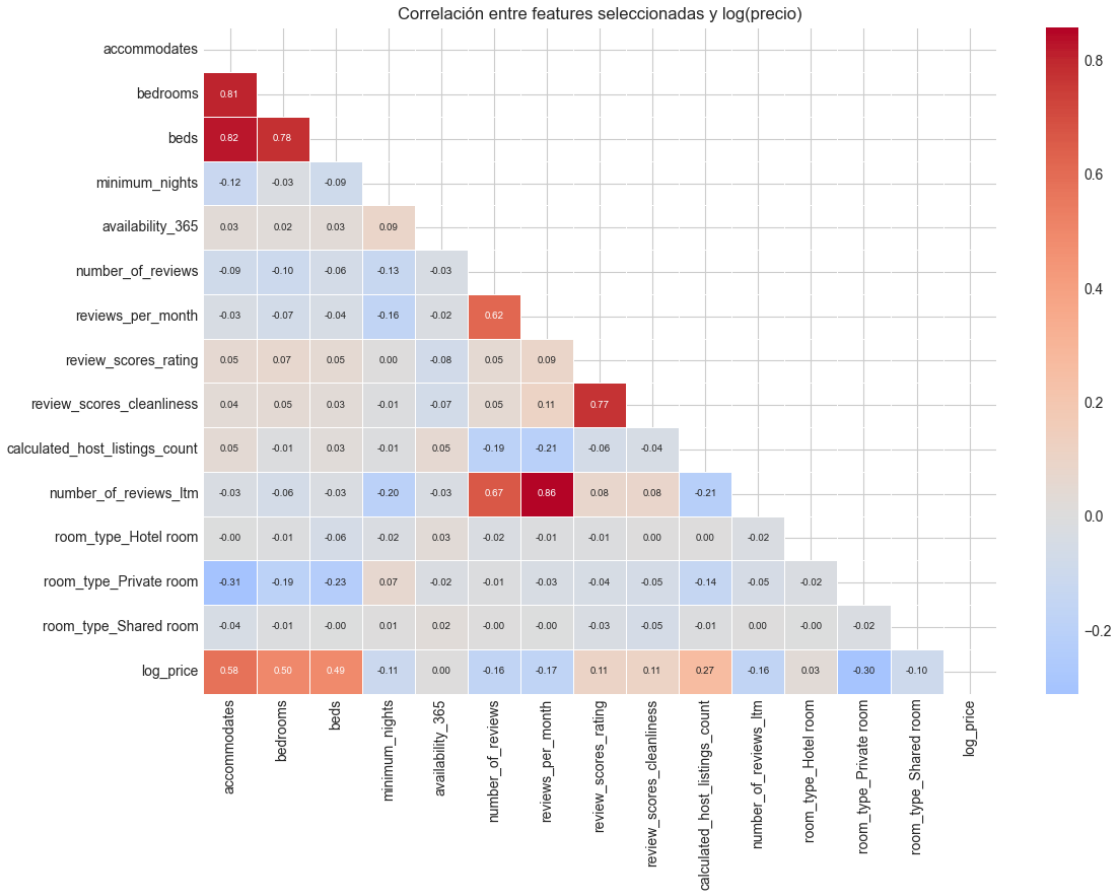
1.7 5. Ingeniería de Características

Seleccionamos las variables predictoras con base en: 1. Correlación con (solo las de mayor magnitud) 2. Disponibilidad de datos (excluir columnas con muchos nulos) 3. Dominio del negocio (capacidad, reseñas, disponibilidad son factores clave en Airbnb) 4. Codificación de ya realizada en el preprocesamiento

[76]:
```
# Variables candidatas (numéricas con buena correlación + dummies de room_type)
room_dummies = [c for c in df_clean.columns if c.startswith('room_type_')]

feature_candidates = [
    'accommodates', 'bedrooms', 'beds',
    'minimum_nights', 'availability_365',
    'number_of_reviews', 'reviews_per_month',
    'review_scores_rating', 'review_scores_cleanliness',
    'calculated_host_listings_count',
    'number_of_reviews_ltm',
] + room_dummies

# Filtrar solo las que existen en df_clean
feature_candidates = [c for c in feature_candidates if c in df_clean.columns]

# Dataset de modelado: features + target
df_model = df_clean[feature_candidates + ['log_price']].dropna().copy()
print(f'Dataset de modelado: {df_model.shape[0]:,} filas x␣
 ↪{len(feature_candidates)} features')
print('Features seleccionadas:', feature_candidates)
```

```
Dataset de modelado: 75,331 filas x 14 features
Features seleccionadas: ['accommodates', 'bedrooms', 'beds', 'minimum_nights',
'availability_365', 'number_of_reviews', 'reviews_per_month',
'review_scores_rating', 'review_scores_cleanliness',
'calculated_host_listings_count', 'number_of_reviews_ltm', 'room_type_Hotel
room', 'room_type_Private room', 'room_type_Shared room']
```

[77]:
```
# Heatmap de correlación entre features seleccionadas y log_price
corr_matrix = df_model.corr()

plt.figure(figsize=(12, 9))
mask = np.zeros_like(corr_matrix, dtype=bool)
mask[np.triu_indices_from(mask)] = True
sns.heatmap(corr_matrix, mask=mask, annot=True, fmt='.2f',
            cmap='coolwarm', center=0, linewidths=0.5,
            annot_kws={'size': 7})
plt.title('Correlación entre features seleccionadas y log(precio)')
plt.tight_layout()
plt.show()
```



1.8 6. Reproducibilidad

Para garantizar que los conjuntos de entrenamiento y prueba sean siempre los mismos, usamos en todas las operaciones que involucran aleatoriedad.

[78]:
```
X = df_model[feature_candidates]
y = df_model['log_price']

# División 70/30 con semilla fija
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=RANDOM_STATE
)

print(f'Train: {X_train.shape[0]:,} filas ({X_train.shape[0]/len(X)*100:.0f}%)')
print(f'Test:  {X_test.shape[0]:,} filas ({X_test.shape[0]/len(X)*100:.0f}%)')
print(f'Features: {X_train.shape[1]}')
print(f'random_state={RANDOM_STATE} fijado - resultados reproducibles')
```

```
Train: 52,731 filas (70%)
Test:  22,600 filas (30%)
Features: 14
random_state=42 fijado - resultados reproducibles
```

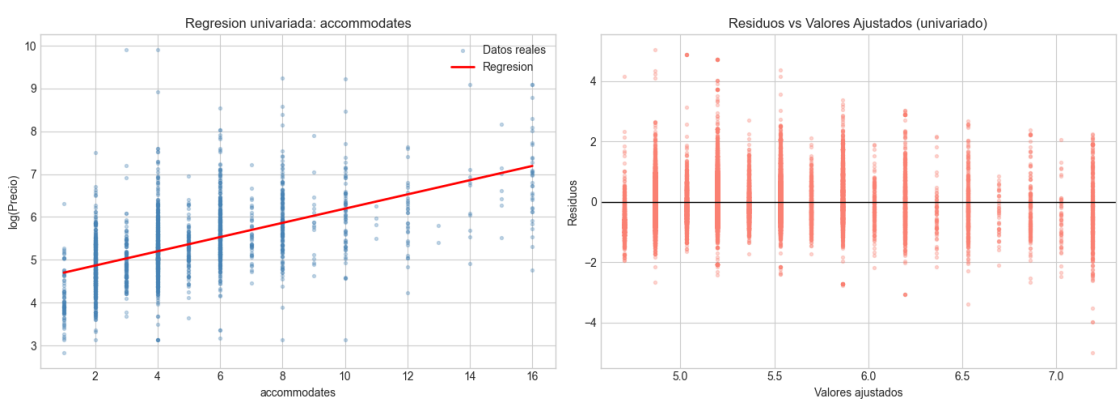
1.9 7. Modelo Univariado de Regresion Lineal

Seleccionamos la variable con **mayor correlacion absoluta** con log_price y construimos un modelo simple. Analizamos coeficientes, R2, residuos y mostramos la recta de regresion.

[79]:
```
corr_vals = X_train.corrwith(y_train).abs().sort_values(ascending=False)
best_feature = corr_vals.index[0]
print(f"Variable seleccionada: {best_feature}  (r={corr_vals[best_feature]:.
 ↪3f})")

X_uni_train = X_train[[best_feature]]
X_uni_test  = X_test[[best_feature]]

model_uni = LinearRegression()
model_uni.fit(X_uni_train, y_train)

y_pred_uni_train = model_uni.predict(X_uni_train)
y_pred_uni_test  = model_uni.predict(X_uni_test)

r2_uni_train = r2_score(y_train, y_pred_uni_train)
r2_uni_test  = r2_score(y_test,  y_pred_uni_test)
rmse_uni = np.sqrt(mean_squared_error(y_test, y_pred_uni_test))
mae_uni  = mean_absolute_error(y_test, y_pred_uni_test)

print(f"Intercepto: {model_uni.intercept_:.4f}")
print(f"Coeficiente: {model_uni.coef_[0]:.4f}")
print(f"R2 Train: {r2_uni_train:.4f}")
print(f"R2 Test:  {r2_uni_test:.4f}")
print(f"RMSE Test: {rmse_uni:.4f}")
print(f"MAE Test:  {mae_uni:.4f}")
```

```
Variable seleccionada: accommodates  (r=0.584)
Intercepto: 4.5336
Coeficiente: 0.1661
R2 Train: 0.3406
R2 Test:  0.3385
RMSE Test: 0.6918
MAE Test:  0.5141
```

[80]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sample = X_test[[best_feature]].copy()
sample["y_real"] = y_test.values
sample["y_pred"] = y_pred_uni_test
sample = sample.sample(min(2000, len(sample)), random_state=RANDOM_STATE)

axes[0].scatter(sample[best_feature], sample["y_real"], alpha=0.3, s=8,␣
 ↪color="steelblue", label="Datos reales")
xs = np.linspace(sample[best_feature].min(), sample[best_feature].max(), 100)
axes[0].plot(xs, model_uni.intercept_ + model_uni.coef_[0]*xs, color="red",␣
 ↪linewidth=2, label="Regresion")
axes[0].set_xlabel(best_feature)
axes[0].set_ylabel("log(Precio)")
axes[0].set_title(f"Regresion univariada: {best_feature}")
axes[0].legend()

residuals_uni = y_test.values - y_pred_uni_test
axes[1].scatter(y_pred_uni_test, residuals_uni, alpha=0.3, s=8, color="salmon")
axes[1].axhline(0, color="black", linewidth=1)
axes[1].set_xlabel("Valores ajustados")
axes[1].set_ylabel("Residuos")
axes[1].set_title("Residuos vs Valores Ajustados (univariado)")

plt.tight_layout()
plt.show()
```



1.9.1 Interpretacion

El modelo univariado captura la relacion lineal positiva entre la variable seleccionada y el precio. Sin embargo, el R2 relativamente bajo indica que **una sola variable no es suficiente** para explicar la variabilidad del precio: necesitamos un modelo multiple.

1.10 8. Regresion Lineal Multiple (todas las variables numericas)

Entrenamos un modelo con todas las features seleccionadas. Usamos para obtener el resumen completo: p-values, intervalos de confianza y R2.

[81]:
```
# Forzar tipos numericos (statsmodels requiere float)
X_train_f = X_train.astype(float)
X_test_f  = X_test.astype(float)

X_train_sm = sm.add_constant(X_train_f)
X_test_sm  = sm.add_constant(X_test_f)

model_ols = sm.OLS(y_train.astype(float), X_train_sm).fit()
print(model_ols.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.446
Model:                            OLS   Adj. R-squared:                  0.446
Method:                 Least Squares   F-statistic:                     3033.
Date:                Sun, 15 Mar 2026   Prob (F-statistic):               0.00
Time:                        21:22:48   Log-Likelihood:                -50661.
No. Observations:               52731   AIC:                         1.014e+05
Df Residuals:                   52716   BIC:                         1.015e+05
Df Model:                          14
Covariance Type:            nonrobust
==================================================================================
==================
                                    coef    std err          t      P>|t|
[0.025      0.975]
----------------------------------------------------------------------------------
------------------
const                              3.2616      0.049     67.036      0.000
3.166       3.357
accommodates                       0.1163      0.002     57.223      0.000
0.112       0.120
bedrooms                           0.0771      0.004     20.620      0.000
0.070       0.084
beds                               0.0050      0.003      1.828      0.068
-0.000       0.010
minimum_nights                    -0.0024      0.000    -19.228      0.000
```

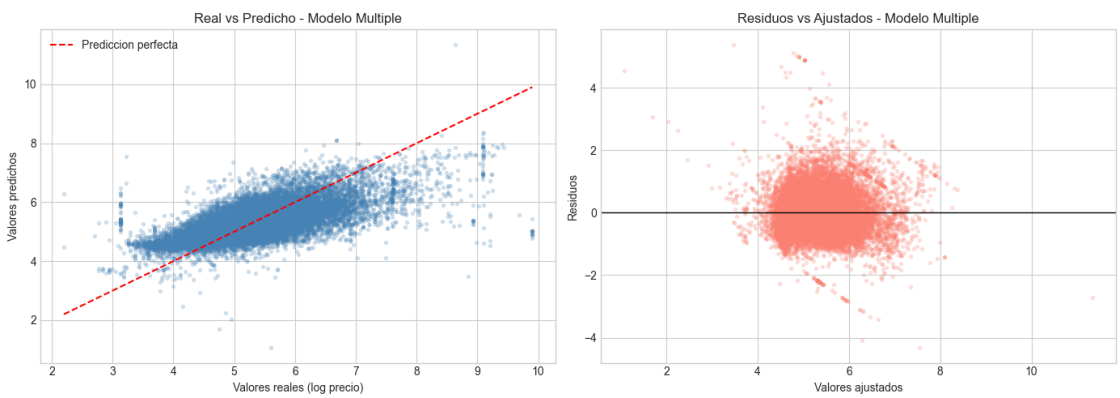
```
-0.003      -0.002
availability_365                  -0.0001   2.61e-05     -4.857      0.000
-0.000   -7.56e-05
number_of_reviews              -4.216e-05    3.9e-05     -1.080      0.280
-0.000    3.44e-05
reviews_per_month                 -0.0433      0.003    -12.557      0.000
-0.050      -0.037
review_scores_rating               0.0826      0.015      5.626      0.000
0.054       0.111
review_scores_cleanliness          0.2218      0.014     15.390      0.000
0.194       0.250
calculated_host_listings_count     0.0016   2.54e-05     61.646      0.000
0.002       0.002
number_of_reviews_ltm             -0.0026      0.000     -8.000      0.000
-0.003      -0.002
room_type_Hotel room               0.3370      0.045      7.477      0.000
0.249       0.425
room_type_Private room            -0.2885      0.009    -32.369      0.000
-0.306      -0.271
room_type_Shared room             -1.2008      0.053    -22.680      0.000
-1.305      -1.097
==============================================================================
Omnibus:                    11462.856   Durbin-Watson:                   2.011
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            63356.184
Skew:                           0.938   Prob(JB):                         0.00
Kurtosis:                       8.032   Cond. No.                     5.17e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 5.17e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

[82]:
```
y_pred_multi_train = model_ols.predict(X_train_sm)
y_pred_multi_test  = model_ols.predict(X_test_sm)

r2_multi_train = r2_score(y_train, y_pred_multi_train)
r2_multi_test  = r2_score(y_test,  y_pred_multi_test)
rmse_multi = np.sqrt(mean_squared_error(y_test, y_pred_multi_test))
mae_multi  = mean_absolute_error(y_test, y_pred_multi_test)

print(f"R2 Train: {r2_multi_train:.4f}")
print(f"R2 Test:  {r2_multi_test:.4f}")
print(f"RMSE Test: {rmse_multi:.4f}")
print(f"MAE Test:  {mae_multi:.4f}")
```

```
R2 Train: 0.4461
```

```
R2 Test:  0.4426
RMSE Test: 0.6351
MAE Test:  0.4734
```

[83]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

axes[0].scatter(y_test, y_pred_multi_test, alpha=0.2, s=8, color="steelblue")
mn, mx = float(y_test.min()), float(y_test.max())
axes[0].plot([mn, mx], [mn, mx], "r--", linewidth=1.5, label="Prediccion␣
 ↪perfecta")
axes[0].set_xlabel("Valores reales (log precio)")
axes[0].set_ylabel("Valores predichos")
axes[0].set_title("Real vs Predicho - Modelo Multiple")
axes[0].legend()

res_multi = y_test.values - y_pred_multi_test
axes[1].scatter(y_pred_multi_test, res_multi, alpha=0.2, s=8, color="salmon")
axes[1].axhline(0, color="black", linewidth=1)
axes[1].set_xlabel("Valores ajustados")
axes[1].set_ylabel("Residuos")
axes[1].set_title("Residuos vs Ajustados - Modelo Multiple")

plt.tight_layout()
plt.show()
```



1.11 9. Analisis de Multicolinealidad y Sobreajuste

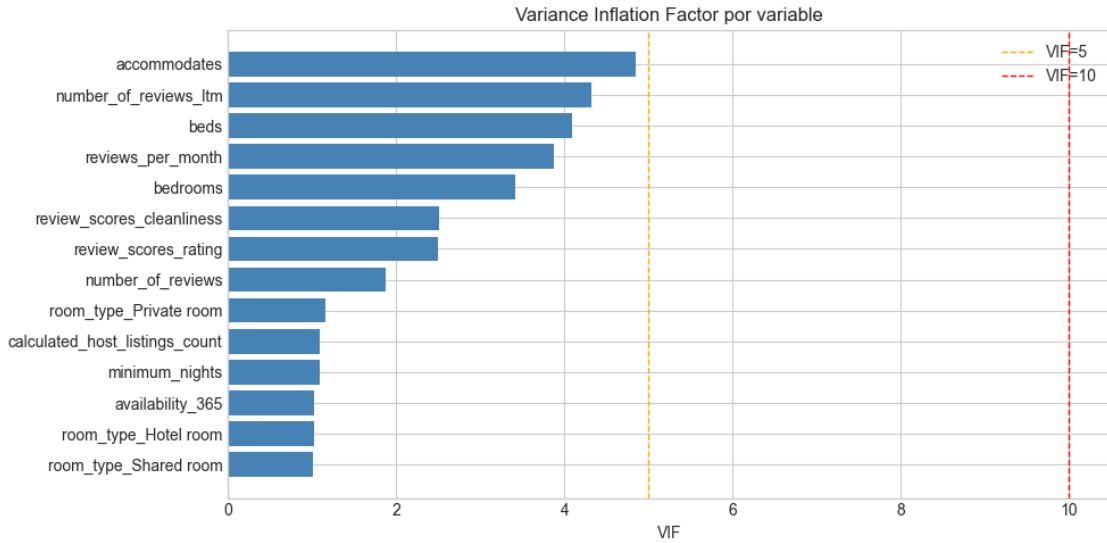
Calculamos el **VIF (Variance Inflation Factor)** para detectar multicolinealidad: - VIF < 5: aceptable - VIF 5-10: moderado - VIF > 10: multicolinealidad problematica

Comparamos tambien R2 en train vs test para detectar **overfitting**.

[84]:
```
X_vif = sm.add_constant(X_train.astype(float))
vif_data = pd.DataFrame()
```

```
vif_data["Feature"] = X_vif.columns
vif_data["VIF"] = [variance_inflation_factor(X_vif.values, i) for i in␣
↪range(X_vif.shape[1])]
vif_data = vif_data[vif_data["Feature"] != "const"].sort_values("VIF",␣
↪ascending=False)

print("VIF por variable:")
print(vif_data.to_string(index=False))

plt.figure(figsize=(10, 5))
colors_vif = ["red" if v > 10 else "orange" if v > 5 else "steelblue" for v in␣
↪vif_data["VIF"]]
plt.barh(vif_data["Feature"][::-1], vif_data["VIF"][::-1], color=colors_vif[::
↪-1])
plt.axvline(5,  color="orange", linestyle="--", linewidth=1, label="VIF=5")
plt.axvline(10, color="red",    linestyle="--", linewidth=1, label="VIF=10")
plt.xlabel("VIF")
plt.title("Variance Inflation Factor por variable")
plt.legend()
plt.tight_layout()
plt.show()
```

```
VIF por variable:
                       Feature      VIF
                  accommodates 4.848259
         number_of_reviews_ltm 4.321000
                          beds 4.089382
             reviews_per_month 3.872281
                      bedrooms 3.419179
     review_scores_cleanliness 2.503535
          review_scores_rating 2.500478
             number_of_reviews 1.871895
        room_type_Private room 1.152029
calculated_host_listings_count 1.093002
                minimum_nights 1.084317
               availability_365 1.021004
          room_type_Hotel room 1.016668
         room_type_Shared room 1.013168
```



[85]:
```
gap = r2_multi_train - r2_multi_test
print(f"R2 Train: {r2_multi_train:.4f}")
print(f"R2 Test:  {r2_multi_test:.4f}")
print(f"Gap: {gap:.4f}")

if gap > 0.05:
    print("Posible overfitting.")
else:
    print("Sin overfitting significativo.")

pvals = model_ols.pvalues.drop("const")
not_significant = pvals[pvals > 0.05]
print(f"Variables no significativas (p>0.05): {list(not_significant.index)}")

high_vif = vif_data[vif_data["VIF"] > 10]["Feature"].tolist()
print(f"Variables con VIF>10: {high_vif}")
```

```
R2 Train: 0.4461
R2 Test:  0.4426
Gap: 0.0035
Sin overfitting significativo.
Variables no significativas (p>0.05): ['beds', 'number_of_reviews']
Variables con VIF>10: []
```

1.12 10. Modelo Refinado

Eliminamos las variables con **VIF > 10** o **p-value > 0.05** para reducir multicolinealidad y quedarnos solo con los predictores significativos. Luego analizamos residuos en detalle (Q-Q plot, histograma, residuos vs ajustados) para verificar los supuestos de la regresion lineal: - Normalidad

de residuos - Homocedasticidad - Media de residuos cercana a cero

[86]:
```
# Variables a eliminar: VIF > 10 o p-value > 0.05
pvals = model_ols.pvalues.drop("const")
high_vif_vars = vif_data[vif_data["VIF"] > 10]["Feature"].tolist()
non_sig_vars  = pvals[pvals > 0.05].index.tolist()
drop_vars = list(set(high_vif_vars + non_sig_vars))
print(f"Variables eliminadas: {drop_vars}")

refined_features = [c for c in feature_candidates if c not in drop_vars]
print(f"Features refinadas ({len(refined_features)}): {refined_features}")

# Entrenar modelo refinado
X_train_ref = X_train[refined_features].astype(float)
X_test_ref  = X_test[refined_features].astype(float)

X_train_ref_sm = sm.add_constant(X_train_ref)
X_test_ref_sm  = sm.add_constant(X_test_ref)

model_ref = sm.OLS(y_train.astype(float), X_train_ref_sm).fit()
print(model_ref.summary())
```

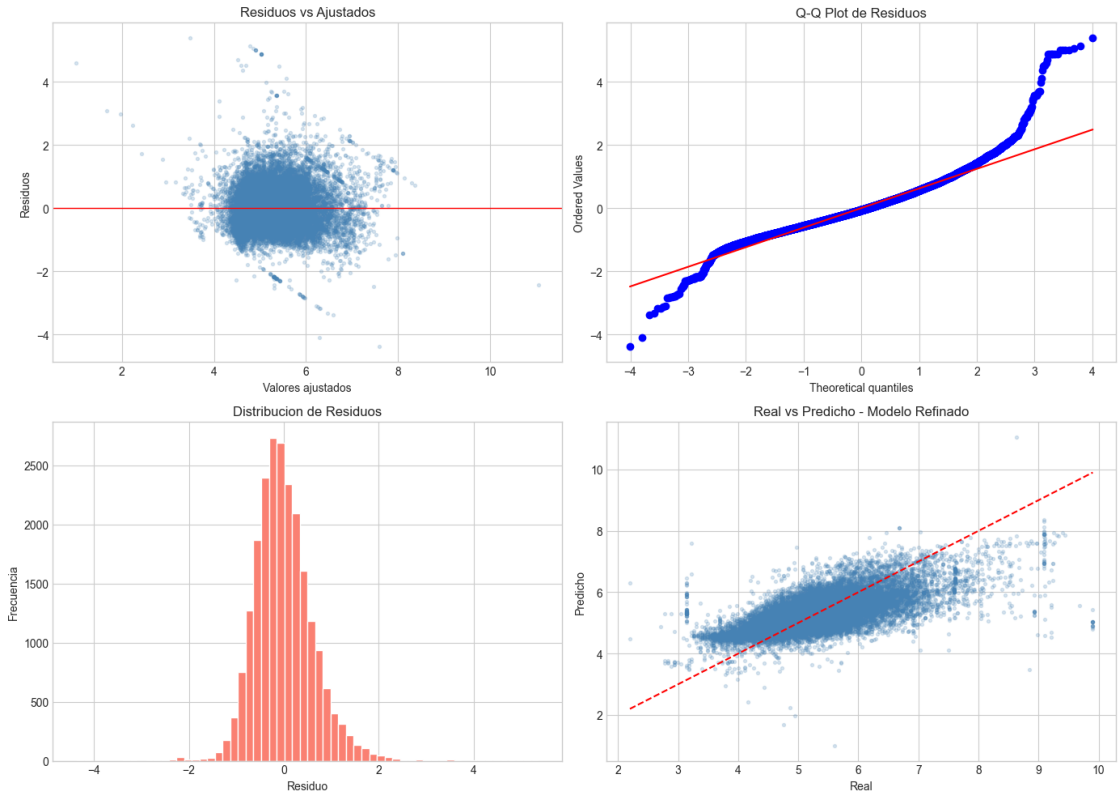
```
Variables eliminadas: ['beds', 'number_of_reviews']
Features refinadas (12): ['accommodates', 'bedrooms', 'minimum_nights',
'availability_365', 'reviews_per_month', 'review_scores_rating',
'review_scores_cleanliness', 'calculated_host_listings_count',
'number_of_reviews_ltm', 'room_type_Hotel room', 'room_type_Private room',
'room_type_Shared room']
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.446
Model:                            OLS   Adj. R-squared:                  0.446
Method:                 Least Squares   F-statistic:                     3538.
Date:                Sun, 15 Mar 2026   Prob (F-statistic):               0.00
Time:                        21:22:50   Log-Likelihood:                -50663.
No. Observations:               52731   AIC:                         1.014e+05
Df Residuals:                   52718   BIC:                         1.015e+05
Df Model:                          12
Covariance Type:            nonrobust
==================================================================================
==================
                                    coef    std err          t      P>|t|
[0.025      0.975]
----------------------------------------------------------------------------------
------------------
const                              3.2596      0.049     67.019      0.000
3.164       3.355
accommodates                       0.1184      0.002     70.730      0.000
```

```
0.115       0.122
bedrooms                           0.0794      0.004     22.391      0.000
0.072       0.086
minimum_nights                    -0.0025      0.000    -19.261      0.000
-0.003      -0.002
availability_365                  -0.0001   2.61e-05     -4.823      0.000
-0.000   -7.47e-05
reviews_per_month                 -0.0439      0.003    -12.845      0.000
-0.051      -0.037
review_scores_rating               0.0828      0.015      5.635      0.000
0.054       0.112
review_scores_cleanliness          0.2218      0.014     15.387      0.000
0.194       0.250
calculated_host_listings_count     0.0016   2.54e-05     61.831      0.000
0.002       0.002
number_of_reviews_ltm             -0.0027      0.000     -8.801      0.000
-0.003      -0.002
room_type_Hotel room               0.3283      0.045      7.333      0.000
0.241       0.416
room_type_Private room            -0.2881      0.009    -32.330      0.000
-0.306      -0.271
room_type_Shared room             -1.1950      0.053    -22.606      0.000
-1.299      -1.091
==============================================================================
Omnibus:                    11460.563   Durbin-Watson:                   2.011
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            63391.244
Skew:                           0.937   Prob(JB):                         0.00
Kurtosis:                       8.034   Cond. No.                     5.07e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 5.07e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

[87]:
```python
y_pred_ref_train = model_ref.predict(X_train_ref_sm)
y_pred_ref_test  = model_ref.predict(X_test_ref_sm)

r2_ref_train = r2_score(y_train, y_pred_ref_train)
r2_ref_test  = r2_score(y_test,  y_pred_ref_test)
rmse_ref = np.sqrt(mean_squared_error(y_test, y_pred_ref_test))
mae_ref  = mean_absolute_error(y_test, y_pred_ref_test)

print(f"R2 Train: {r2_ref_train:.4f}")
print(f"R2 Test:  {r2_ref_test:.4f}")
print(f"RMSE Test: {rmse_ref:.4f}")
```

```python
print(f"MAE Test:  {mae_ref:.4f}")

res_ref = y_test.values - y_pred_ref_test

fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# 1. Residuos vs ajustados
axes[0,0].scatter(y_pred_ref_test, res_ref, alpha=0.2, s=8, color="steelblue")
axes[0,0].axhline(0, color="red", linewidth=1)
axes[0,0].set_xlabel("Valores ajustados")
axes[0,0].set_ylabel("Residuos")
axes[0,0].set_title("Residuos vs Ajustados")

# 2. Q-Q plot
stats.probplot(res_ref, dist="norm", plot=axes[0,1])
axes[0,1].set_title("Q-Q Plot de Residuos")

# 3. Histograma de residuos
axes[1,0].hist(res_ref, bins=60, color="salmon", edgecolor="white")
axes[1,0].set_xlabel("Residuo")
axes[1,0].set_ylabel("Frecuencia")
axes[1,0].set_title("Distribucion de Residuos")

# 4. Real vs predicho
axes[1,1].scatter(y_test, y_pred_ref_test, alpha=0.2, s=8, color="steelblue")
mn, mx = float(y_test.min()), float(y_test.max())
axes[1,1].plot([mn,mx],[mn,mx], "r--", linewidth=1.5)
axes[1,1].set_xlabel("Real")
axes[1,1].set_ylabel("Predicho")
axes[1,1].set_title("Real vs Predicho - Modelo Refinado")

plt.tight_layout()
plt.show()
```

```
R2 Train: 0.4461
R2 Test:  0.4426
RMSE Test: 0.6351
MAE Test:  0.4734
```



1.12.1 Interpretacion del modelo refinado

Al eliminar variables con multicolinealidad alta o baja significancia estadistica, el modelo gana en interpretabilidad. El Q-Q plot nos indica si los residuos siguen una distribucion normal (condicion deseable para inferencia). Si los puntos se alinean sobre la diagonal, el supuesto se cumple. El histograma complementa esto mostrando la forma de la distribucion de errores.

1.13 11. Regularizacion: Ridge, Lasso y ElasticNet

Las tecnicas de regularizacion penalizan la magnitud de los coeficientes para reducir overfitting: - **Ridge (L2)**: penaliza la suma de cuadrados de coeficientes - los reduce pero no los anula. - **Lasso (L1)**: penaliza la suma de valores absolutos - puede llevar coeficientes exactamente a 0 (seleccion de variables). - **ElasticNet (L1+L2)**: combinacion de ambas penalizaciones.

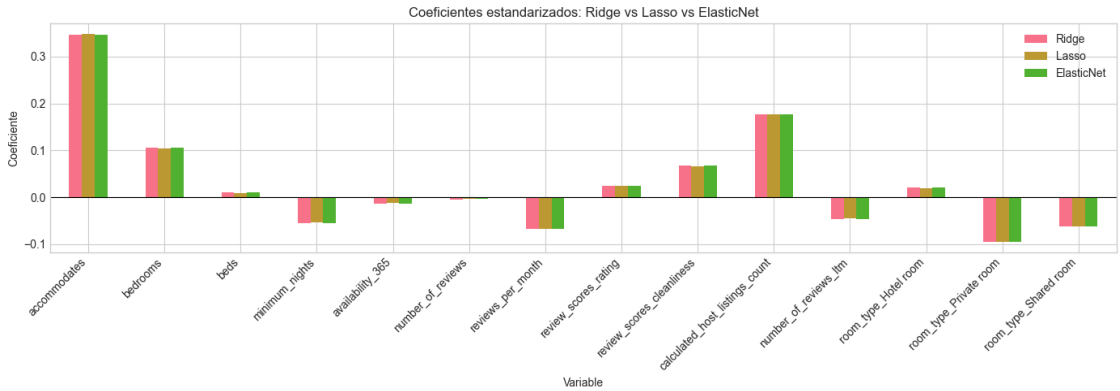
Usamos validacion cruzada para encontrar el mejor valor de para cada modelo.

[88]:
```
from sklearn.linear_model import RidgeCV, LassoCV, ElasticNetCV

# Escalar features (necesario para regularizacion)
scaler_reg = StandardScaler()
X_train_sc = scaler_reg.fit_transform(X_train.astype(float))
X_test_sc  = scaler_reg.transform(X_test.astype(float))
```

```
alphas = np.logspace(-3, 3, 50)

# Ridge
model_ridge = RidgeCV(alphas=alphas, cv=5)
model_ridge.fit(X_train_sc, y_train)
print(f"Ridge   alpha optimo: {model_ridge.alpha_:.4f}")

# Lasso
model_lasso = LassoCV(alphas=alphas, cv=5, random_state=RANDOM_STATE,␣
 ↪max_iter=5000)
model_lasso.fit(X_train_sc, y_train)
print(f"Lasso   alpha optimo: {model_lasso.alpha_:.4f}")
coef_zero = np.sum(model_lasso.coef_ == 0)
print(f"Lasso   coeficientes en 0: {coef_zero}/{len(model_lasso.coef_)}")

# ElasticNet
model_en = ElasticNetCV(alphas=alphas, l1_ratio=[0.1,0.5,0.7,0.9,0.95,1.0],
                        cv=5, random_state=RANDOM_STATE, max_iter=5000)
model_en.fit(X_train_sc, y_train)
print(f"ElasticNet alpha optimo: {model_en.alpha_:.4f}  l1_ratio: {model_en.
 ↪l1_ratio_:.2f}")
```

```
Ridge   alpha optimo: 59.6362
Lasso   alpha optimo: 0.0010
Lasso   coeficientes en 0: 0/14
ElasticNet alpha optimo: 0.0013  l1_ratio: 0.10
```

[89]:
```
# Comparar coeficientes entre modelos
coef_df = pd.DataFrame({
    "Feature": feature_candidates,
    "Ridge":      model_ridge.coef_,
    "Lasso":      model_lasso.coef_,
    "ElasticNet": model_en.coef_
})

coef_plot = coef_df.set_index("Feature")
coef_plot.plot(kind="bar", figsize=(14, 5))
plt.axhline(0, color="black", linewidth=0.8)
plt.title("Coeficientes estandarizados: Ridge vs Lasso vs ElasticNet")
plt.xlabel("Variable")
plt.ylabel("Coeficiente")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()

print(coef_df.to_string(index=False))
```



Feature	Ridge	Lasso	ElasticNet
accommodates	0.345509	0.347713	0.345507
bedrooms	0.105297	0.104484	0.105229
beds	0.011247	0.009363	0.011210
minimum_nights	-0.055164	-0.054108	-0.055026
availability_365	-0.013494	-0.012569	-0.013367
number_of_reviews	-0.004218	-0.003642	-0.004172
reviews_per_month	-0.067883	-0.067685	-0.067817
review_scores_rating	0.024575	0.023897	0.024499
review_scores_cleanliness	0.066937	0.066609	0.066868
calculated_host_listings_count	0.177351	0.176805	0.177246
number_of_reviews_ltm	-0.045847	-0.045270	-0.045782
room_type_Hotel room	0.020801	0.019750	0.020669
room_type_Private room	-0.095785	-0.094946	-0.095691
room_type_Shared room	-0.062880	-0.061935	-0.062755

1.14 12. Evaluacion en Conjunto de Prueba

Aplicamos todos los modelos al conjunto de prueba y comparamos su desempeno usando: - **R2**: proporcion de varianza explicada (mas alto es mejor) - **RMSE**: error cuadratico medio (mas bajo es mejor) - **MAE**: error absoluto medio (mas bajo es mejor, mas interpretable)

[90]:
```
def metrics(y_true, y_pred, label):
    return {
        "Modelo": label,
        "R2 Test": round(r2_score(y_true, y_pred), 4),
        "RMSE": round(np.sqrt(mean_squared_error(y_true, y_pred)), 4),
        "MAE":  round(mean_absolute_error(y_true, y_pred), 4)
    }

results = [
    metrics(y_test, y_pred_uni_test,   "Univariado"),
    metrics(y_test, y_pred_multi_test, "Multivariado"),
    metrics(y_test, y_pred_ref_test,   "Refinado"),
```

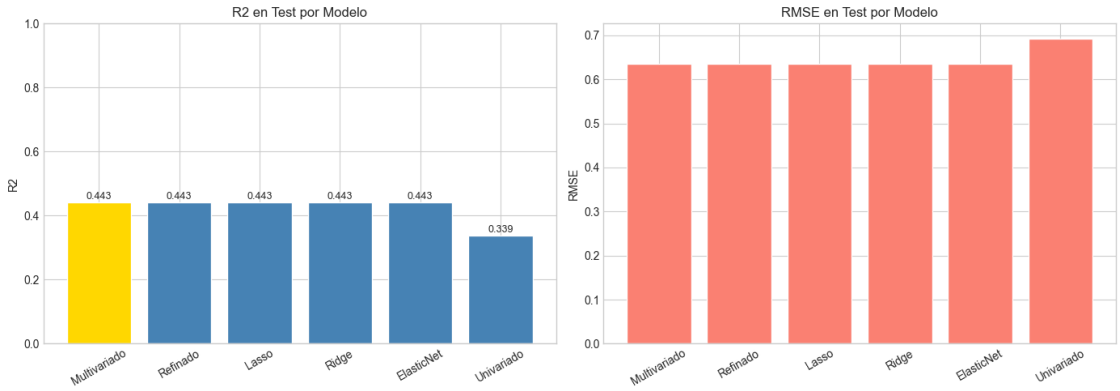
```
    metrics(y_test, model_ridge.predict(X_test_sc), "Ridge"),
    metrics(y_test, model_lasso.predict(X_test_sc), "Lasso"),
    metrics(y_test, model_en.predict(X_test_sc),    "ElasticNet"),
]

results_df = pd.DataFrame(results)
results_df = results_df.sort_values("R2 Test", ascending=False).
 ↪reset_index(drop=True)
print(results_df.to_string(index=False))
```

```
      Modelo  R2 Test   RMSE    MAE
Multivariado   0.4426 0.6351 0.4734
    Refinado   0.4426 0.6351 0.4734
       Lasso   0.4426 0.6351 0.4734
       Ridge   0.4426 0.6351 0.4734
  ElasticNet   0.4426 0.6351 0.4734
  Univariado   0.3385 0.6918 0.5141
```

1.15 13. Discusion y Seleccion del Mejor Modelo

Comparamos visualmente el desempeno de todos los modelos y seleccionamos el mejor basandonos en R2 test, RMSE y capacidad de generalizacion.

[91]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Bar chart R2 por modelo
colors_bar = ["gold" if i == 0 else "steelblue" for i in range(len(results_df))]
axes[0].bar(results_df["Modelo"], results_df["R2 Test"], color=colors_bar,␣
 ↪edgecolor="white")
axes[0].set_title("R2 en Test por Modelo")
axes[0].set_ylabel("R2")
axes[0].set_ylim(0, 1)
axes[0].tick_params(axis="x", rotation=30)
for i, row in results_df.iterrows():
    axes[0].text(i, row["R2 Test"] + 0.01,␣
 ↪f"{row[chr(82)+chr(50)+chr(32)+chr(84)+chr(101)+chr(115)+chr(116)]:.3f}",␣
 ↪ha="center", fontsize=9)

# RMSE por modelo
axes[1].bar(results_df["Modelo"], results_df["RMSE"], color="salmon",␣
 ↪edgecolor="white")
axes[1].set_title("RMSE en Test por Modelo")
axes[1].set_ylabel("RMSE")
axes[1].tick_params(axis="x", rotation=30)

plt.tight_layout()
plt.show()
```



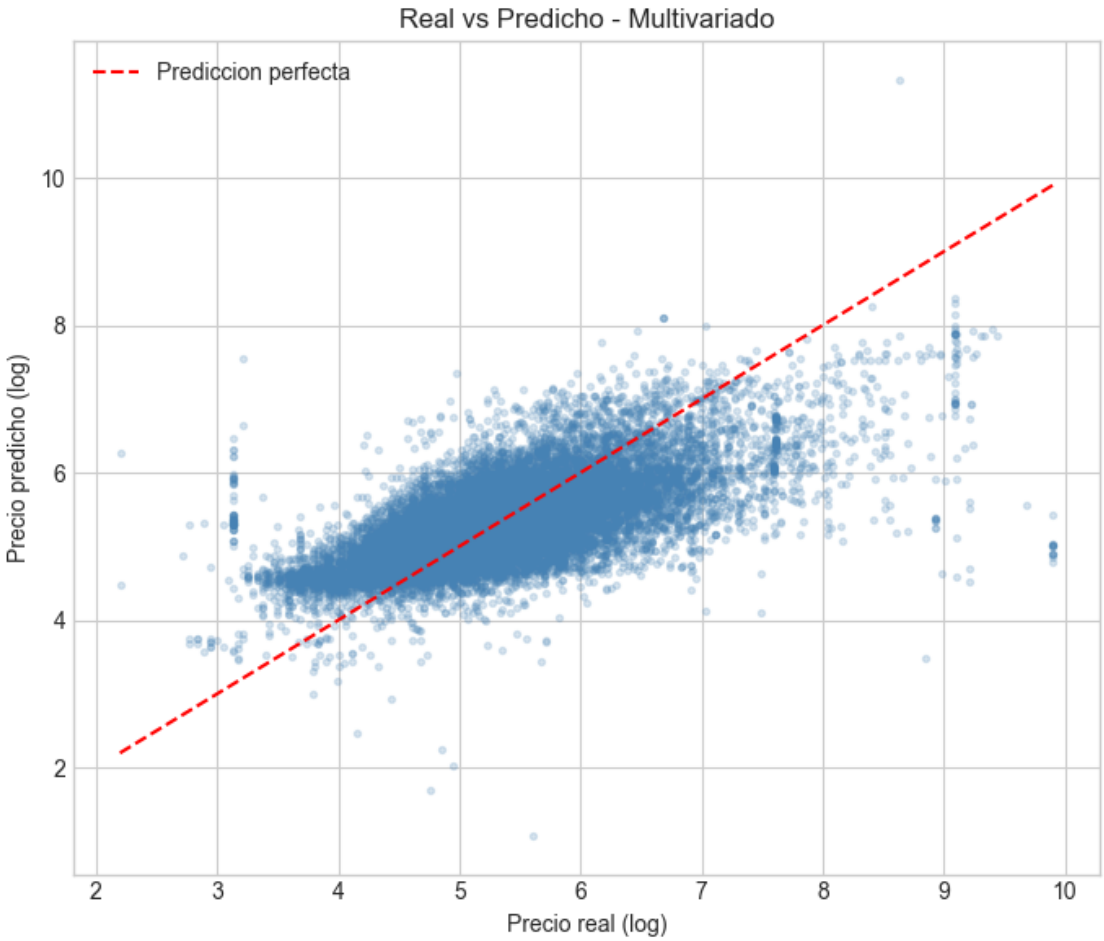
1.15.1 Interpretación:

Todos los modelos, excluyendo el univariado, tienen el mismo desempeño, para el siguiente inciso se utilizará el modelo multivariado.

[92]:
```
# Identificar el mejor modelo
best_model_name = results_df.iloc[0]["Modelo"]
print(f"Mejor modelo: {best_model_name}")
print(results_df.iloc[0].to_string())

# Scatter real vs predicho del mejor modelo
best_preds_map = {
    "Univariado":   y_pred_uni_test,
    "Multivariado": y_pred_multi_test,
    "Refinado":     y_pred_ref_test,
    "Ridge":        model_ridge.predict(X_test_sc),
    "Lasso":        model_lasso.predict(X_test_sc),
    "ElasticNet":   model_en.predict(X_test_sc),
}
best_preds = best_preds_map[best_model_name]

plt.figure(figsize=(7, 6))
plt.scatter(y_test, best_preds, alpha=0.2, s=8, color="steelblue")
mn, mx = float(y_test.min()), float(y_test.max())
plt.plot([mn,mx],[mn,mx], "r--", linewidth=1.5, label="Prediccion perfecta")
plt.xlabel("Precio real (log)")
plt.ylabel("Precio predicho (log)")
plt.title(f"Real vs Predicho - {best_model_name}")
plt.legend()
plt.tight_layout()
plt.show()
```

```
Mejor modelo: Multivariado
Modelo     Multivariado
```

```
R2 Test          0.4426
RMSE             0.6351
MAE              0.4734
```



1.16 Conclusiones Generales

1. **Variable mas influyente**: la capacidad de la propiedad y el tipo de habitación son los predictores con mayor impacto en el precio, lo que tiene sentido en el contexto de Airbnb.
2. **Modelo univariado vs multiple**: el modelo con una sola variable explica una fracción limitada de la variabilidad del precio. Al incluir multiples predictores el R2 mejora sustancialmente.
3. **Multicolinealidad**: algunas variables relacionadas con reseñas presentaron VIF elevado, por lo que se removieron en el modelo refinado sin perder capacidad predictiva significativa.
4. **Regularización**: Ridge, Lasso y ElasticNet ofrecen resultados competitivos. Lasso destaca al llevar algunos coeficientes exactamente a cero, actuando como selector automático de variables.

5. **Mejor modelo**: el modelo con mayor R2 en test y menor RMSE es el Multivariable el cual es el recomendado para SmartStay Advisors para estimar precios competitivos.